

# Direct and Inverse Proportion

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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## Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

**1** [1 mark]

One pen costs 40p.

How much do 5 pens cost?

1 The cost is directly proportional to the number of pens, so multiply.

2  $5 \times 40\text{p} = 200\text{p}$

**Answer:** £2.00

**2** [2 marks]

4 identical books cost £26.

Work out the cost of 7 of these books.

1 Find the cost of one book:  $26 \div 4 = £6.50$

2 7 books cost  $7 \times 6.50 = £45.50$

**Answer:** £45.50

**3** [2 marks]

3 workers can build a wall in 12 days.

How long would 6 workers, all working at the same rate, take?

1 More workers means less time — this is inverse proportion.

2 Doubling the number of workers halves the time.

3  $12 \div 2 = 6$  days

**Answer:** 6 days

**4** [2 marks]

200 g of cheese costs £3.20

Work out the cost of 350 g of the same cheese.

1 Cost of 1 g =  $320\text{p} \div 200 = 1.6\text{p}$

2 350 g costs  $350 \times 1.6 = 560\text{p} = £5.60$

**Answer:** £5.60

5

[3 marks]

$y$  is directly proportional to  $x$ . When  $x = 4$ ,  $y = 20$

(a) Work out  $y$  when  $x = 7$

(b) Work out  $x$  when  $y = 45$

1 Direct proportion means  $y = kx$  for some fixed number  $k$ .

2  $20 = k \times 4$ , so  $k = 5$  and  $y = 5x$

3 (a)  $y = 5 \times 7 = 35$

4 (b)  $45 = 5x$ , so  $x = 9$

**Answer:** (a) 35 (b) 9

6

[3 marks]

5 machines fill 800 bottles in an hour.

How many bottles would 8 such machines fill in an hour?

1 Find the rate for one machine:  $800 \div 5 = 160$  bottles per hour.

2 8 machines fill  $8 \times 160 = 1280$  bottles

3 More machines means more bottles, so this is direct proportion.

**Answer:** 1280 bottles

7

[3 marks]

6 people take 10 hours to paint a fence.

How long would 4 people take, working at the same rate?

1 Work out the total work in person-hours:  $6 \times 10 = 60$  person-hours.

2 That total does not change, however many people work.

3 Time for 4 people =  $60 \div 4 = 15$  hours

4 Sensible check: fewer people means more time ✓

**Answer:** 15 hours

8

[3 marks]

A recipe for 6 people uses 900 g of pasta.

How much pasta is needed for 10 people?

1 Pasta for one person =  $900 \div 6 = 150$  g

2 For 10 people:  $10 \times 150 = 1500$  g

3 That is 1.5 kg.

**Answer:** 1500 g (1.5 kg)

9

[4 marks]

$y$  is directly proportional to  $x$ . When  $x = 3$ ,  $y = 21$

- (a) Find the value of the constant of proportionality.
- (b) Write a formula for  $y$  in terms of  $x$ .
- (c) Work out  $y$  when  $x = 11$
- (d) Write down where the graph of  $y$  against  $x$  crosses the  $y$ -axis.

1 (a)  $y = kx$ , so  $21 = 3k$  and  $k = 7$

2 (b)  $y = 7x$

3 (c)  $y = 7 \times 11 = 77$

4 (d) When  $x = 0$ ,  $y = 0$ , so the graph passes through the origin  $(0, 0)$ .

5 A direct proportion graph is always a straight line through the origin.

**Answer:** (a) 7 (b)  $y = 7x$  (c) 77 (d)  $(0, 0)$

10

[4 marks]

For each pair of quantities, state whether they are in direct proportion or inverse proportion.

- (a) The number of workers and the time to finish a fixed job.
- (b) The number of litres of petrol bought and the total cost.
- (c) The average speed and the time taken for a fixed journey.
- (d) The number of identical tins and their total mass.

1 (a) More workers means less time, so inverse proportion.

2 (b) More petrol means more cost, at a fixed price per litre, so direct proportion.

3 (c) A higher speed means a shorter time for the same distance, so inverse proportion.

4 (d) More tins means more mass, so direct proportion.

**Answer:** (a) inverse (b) direct (c) inverse (d) direct

## Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

$y$  is directly proportional to  $x$ , and  $y = 12$  when  $x = 3$

Find a formula for  $y$  in terms of  $x$ .

1 Write  $y = kx$

2  $12 = 3k$ , so  $k = 4$

**Answer:**  $y = 4x$

2

[2 marks]

$y$  is inversely proportional to  $x$ , and  $y = 5$  when  $x = 4$

Find a formula for  $y$  in terms of  $x$ .

1 Write  $y = \frac{k}{x}$

2  $5 = \frac{k}{4}$ , so  $k = 20$

**Answer:**  $y = \frac{20}{x}$

3

[3 marks]

$y$  is proportional to  $x^2$ . When  $x = 3$ ,  $y = 45$

Work out  $y$  when  $x = 5$

1 Write  $y = kx^2$

2  $45 = k \times 9$ , so  $k = 5$  and  $y = 5x^2$

3  $y = 5 \times 5^2 = 5 \times 25$

**Answer:** 125

4

[3 marks]

$y$  is proportional to  $\sqrt{x}$ . When  $x = 16$ ,  $y = 12$

Work out  $y$  when  $x = 25$

1 Write  $y = k\sqrt{x}$

2  $\sqrt{16} = 4$ , so  $12 = 4k$  and  $k = 3$

3  $y = 3\sqrt{25} = 3 \times 5$

**Answer:** 15

5

[4 marks]

$y$  is inversely proportional to  $x^2$ . When  $x = 2$ ,  $y = 9$

(a) Find a formula for  $y$  in terms of  $x$ .

(b) Work out  $y$  when  $x = 6$

(c) Work out the positive value of  $x$  when  $y = 4$

1 (a)  $y = \frac{k}{x^2}$ , so  $9 = \frac{k}{4}$  and  $k = 36$

2  $y = \frac{36}{x^2}$

3 (b)  $y = \frac{36}{36} = 1$

4 (c)  $4 = \frac{36}{x^2}$ , so  $4x^2 = 36$

5  $x^2 = 9$ , so the positive value is  $x = 3$

**Answer:** (a)  $y = \frac{36}{x^2}$  (b) 1 (c) 3

6

[4 marks]

The force  $F$  newtons between two magnets is inversely proportional to the square of the distance  $d$  cm between them.

When  $d = 4$ ,  $F = 5$

(a) Find a formula for  $F$  in terms of  $d$ .

(b) Work out  $F$  when  $d = 10$

(c) Work out  $d$  when  $F = 20$

1

$$(a) F = \frac{k}{d^2}, \text{ so } 5 = \frac{k}{16} \text{ and } k = 80$$

2

$$F = \frac{80}{d^2}$$

3

$$(b) F = \frac{80}{100} = 0.8 \text{ N}$$

4

$$(c) 20 = \frac{80}{d^2}, \text{ so } 20d^2 = 80$$

5

$$d^2 = 4, \text{ so } d = 2 \text{ cm (a distance must be positive).}$$

**Answer:** (a)  $F = \frac{80}{d^2}$  (b) 0.8 N (c) 2 cm

7

[4 marks]

(a)  $y$  is proportional to  $x^3$ . What happens to  $y$  if  $x$  is doubled?

(b)  $y$  is inversely proportional to  $x^2$ . What happens to  $y$  if  $x$  is trebled?

1

$$(a) y = kx^3. \text{ Replacing } x \text{ by } 2x \text{ gives } k(2x)^3 = 8kx^3$$

2

So  $y$  is multiplied by 8.

3

$$(b) y = \frac{k}{x^2}. \text{ Replacing } x \text{ by } 3x \text{ gives } \frac{k}{9x^2}$$

4

So  $y$  is divided by 9.

**Answer:** (a) multiplied by 8 (b) divided by 9

8

[4 marks]

The time  $T$  days to build a wall is inversely proportional to the number of workers  $n$ .

6 workers take 15 days.

- Find a formula for  $T$  in terms of  $n$ .
- How long would 10 workers take?
- How many workers are needed to finish in 5 days?

1 (a)  $T = \frac{k}{n}$ , so  $15 = \frac{k}{6}$  and  $k = 90$

2  $T = \frac{90}{n}$

3 (b)  $T = \frac{90}{10} = 9$  days

4 (c)  $5 = \frac{90}{n}$ , so  $5n = 90$

5  $n = 18$  workers

**Answer:** (a)  $T = \frac{90}{n}$  (b) 9 days (c) 18 workers

9

[4 marks]

$y$  is directly proportional to  $x^2$  and inversely proportional to  $z$ .

When  $x = 2$  and  $z = 4$ ,  $y = 3$

Work out  $y$  when  $x = 4$  and  $z = 6$

1 Combine both statements:  $y = \frac{kx^2}{z}$

2 Substitute the known values:  $3 = \frac{k \times 4}{4} = k$

3 So  $k = 3$  and  $y = \frac{3x^2}{z}$

4  $y = \frac{3 \times 16}{6} = \frac{48}{6} = 8$

**Answer:** 8

The distance  $d$  metres needed to stop a car is proportional to the square of its speed  $v$  mph.  
At 30 mph the stopping distance is 45 m.

- (a) Find a formula for  $d$  in terms of  $v$ .
- (b) Work out the stopping distance at 60 mph.
- (c) Explain the effect on the stopping distance of doubling the speed.
- (d) Find the speed at which the stopping distance is 80 m.

1 (a)  $d = kv^2$ , so  $45 = k \times 900$

2  $k = \frac{45}{900} = 0.05$ , giving  $d = 0.05v^2$

3 (b)  $d = 0.05 \times 60^2 = 0.05 \times 3600 = 180$  m

4 (c) Doubling  $v$  multiplies  $v^2$  by 4, so the stopping distance becomes four times as long (45 m  $\rightarrow$  180 m confirms this).

5 (d)  $80 = 0.05v^2$ , so  $v^2 = 1600$

6  $v = 40$  mph

**Answer:** (a)  $d = 0.05v^2$  (b) 180 m (c) it becomes 4 times as long (d) 40 mph